Multiple Probabilities of Direction: A Rationale for Prior-odds Adjustment

Margherita Calderan[1], Filippo Gambarota[2], Enrico Toffalini[1], and Gianmarco Altoè[2]

[1]Department of General Psychology, University of Padova

[2]Department of Developmental Psychology and Socialization, University of Padova

Author Note

Margherita Calderan https://orcid.org/0009-0005-5668-5162

Filippo Gambarota https://orcid.org/0000-0002-6666-1747

Enrico Toffalini https://orcid.org/0000-0002-1404-5133

Gianmarco Altoè https://orcid.org/0000-0003-1254-9528

Correspondence concerning this article should be addressed to Margherita Calderan, Department of General Psychology, University of Padova, Via Venezia 8, Padova 35131, Italy, Email: margherita.calderan@unipd.it

Abstract

In psychological research, Bayesian analyses are often summarized using posterior indices and then interpreted with conventional cutoffs across many parameters. The probability of direction (*pd*) is one such index, defined as the posterior probability that a parameter is greater or smaller than a null value. When *pd* is thresholded and used repeatedly, multiplicity becomes consequential but is rarely handled explicitly. This paper examines the operating characteristics of *pd*-based decisions under repeated use, leveraging its correspondence with the frequentist *p*-value in common settings. Simulations show that extreme *pd* values can occur by chance and that misleading conclusions become more likely as the number of tests increases. We then express pd in terms of posterior odds and propose a principled prior-odds adjustment following Jeffreys' and Westfall's prior-probability approach to multiplicity. We show that a global prior probability that all hypotheses are null can be converted into per-test adjustments that preserve nominal error rates while remaining interpretable in Bayesian terms. An accompanying R package implements the proposed prior-odds correction for routine use in applied settings.

Keywords: probability of direction, multiple testing, prior-odds

Multiple Probabilities of Direction: A Rationale for Prior-odds Adjustment

Over the past two decades, the proportion of articles reporting Bayesian analyses in psychology has increased considerably (Jevremov & Pajić, 2024; Pfadt et al., 2025; Van De Schoot et al., 2017). Unfortunately, this expansion has been accompanied by practices reminiscent of the "null ritual" in frequentist statistics, where widespread use was associated with automatic application rather than principled reasoning (Cohen, 1994; Gigerenzer, 2004). For instance, researchers sometimes treat Bayes Factor thresholds (e.g., BF > 3) as dichotomous decision rules or claim that an effect "exists" because the posterior probability of a positive effect exceeds a conventional benchmark such as 95% (Heck et al., 2023; Hoijtink et al., 2016; Kelter, 2023; Kruschke, 2018; Tendeiro et al., 2024; Tendeiro & Kiers, 2019). While such summaries are convenient, their use as default cutoffs encourages all-or-none interpretations. When these binary judgments are applied repeatedly across many parameters, the issue of multiplicity arises.

In the frequentist framework, multiplicity is typically framed as Type I error (false-positive) problem: if each individual test is conducted at level α, the probability of at least one false positive increases with the number of tests. Classical approaches therefore control either the family-wise error rate (the probability of at least one Type I error) or the false discovery rate (the expected proportion of false positives among rejected hypotheses) (Benjamini & Hochberg, 1995; Westfall & Young, 1993). Additionally, when researchers examine many parameters but selectively report only the most extreme estimates (i.e., those that reach statistical significance), selection effects can produce exaggerated magnitudes, a problem known as Type M error (Altoè et al., 2020; Gelman & Tuerlinckx, 2000; Zwet & Cator, 2021). Bayesian shrinkage methods address these distortions by pulling extreme estimates toward more plausible values. Such shrinkage can arise through hierarchical models with partial pooling (Berry & Hochberg, 1999; Gelman et al., 2012, 2013; Kruschke, 2011) and/or through informative priors that reduce overestimation in low signal-to-noise settings (Dawid, 1994; Zwet & Gelman, 2022). Shrinkage reduces Type M errors, but it does not, on its own, provide a guarantee about the error properties of a repeated decision rule applied across many parameters. In particular, if the same cutoff is applied to many

parameters (e.g., declaring an effect whenever posterior probability exceeds 95%), the chance of at least one spurious finding can remain inflated, even if shrinkage reduces it. Bayesian approaches that directly target this issue typically rely on either simultaneous credible statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020; **?**) or priors on the hypothesis/model space that scale with the number of tested effects (Jeffreys, 1938; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997). Nevertheless, in psychological research shrinkage is often treated as sufficient for multiplicity control (Chen et al., 2020; Erčulj & Štrumbelj, 2015; Hogeveen et al., 2022; Limbachia et al., 2021; Mathar et al., 2022).

Other justifications invokes the subjectivist interpretation of Bayesian inference, which treats posterior probabilities as rational degrees of personal belief (Edwards et al., 1963; Lindley, 1972; Rozeboom, 1960). However, this interpretation fits poorly with common psychological practice, where researchers typically apply default software priors (often weakly informative) rather than priors encoding substantive domain knowledge (Smid & Winter, 2020; Van De Schoot et al., 2017). When these priors are combined with sufficient data, Bayesian and frequentist procedures often converge to similar conclusions (Albers et al., 2018; Bernardo & Smith, 1994; Gelman et al., 2013).

Given this convergence and the prevailing practices in psychological research, we aim to evaluate the frequentist operating characteristics of Bayesian decision rules when examining many parameters. Specifically, we focus on the Probability of Direction (*pd*), a recently proposed and increasingly adopted Bayesian index that can be explicitly linked to the frequentist *p*-value in common settings (Berger & Sellke, 1987; Casella & Berger, 1987; Makowski et al., 2019; Marsman & Wagenmakers, 2017) [more **REFs**]. This connection allows for straightforward evaluation of *pd*-based decision rules under repeated use, enabling direct comparison between Bayesian and frequentist approaches to multiplicity control. We do not claim to offer a definitive solution; instead, we propose a transparent, practical framework that encourages more deliberate reporting and highlights the challenges of inference across many parameters.

The paper proceeds in four steps. First, we introduce *pd* and make its connection to the

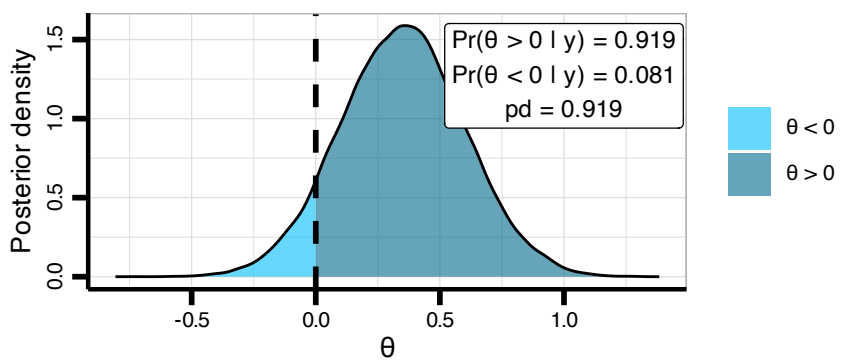
frequentist p-value explicit. Second, we characterize the null behavior of *pd* and, via simulations, examine the multiple-testing properties of *pd*-thresholded decisions, with a focus on family-wise error rate. Third, we express *pd* in terms of posterior odds, relate these odds to directional Bayes factors, and clarify the role of prior odds. Finally, building on Jeffreys' global-null formulation (Jeffreys, 1938) and the Bayesian justification for Bonferroni-style multiplicity adjustment developed by Westfall, Johnson, and Utts (Westfall et al., 1997), we show how these prior-odds ideas extend naturally to *pd*. In the discussion, we offer practical guidance for implementation and reporting.

Probability of Direction

The probability of direction (*pd*) is a descriptive posterior index that quantifies how strongly the posterior distribution favors one direction of an effect relative to a reference (θ_{null}) value (Makowski et al., 2019). For a continuous posterior, it can be defined as the larger of the two one-sided posterior probabilities (Figure 1): $pd = \max(\Pr(\theta < \theta_{null} \mid y),\ \Pr(\theta > \theta_{null} \mid y))$, and it ranges from 0.50 (no directional preference) to 1 (all on one side).

Figure 1

Posterior distribution with directional posterior mass on either side of θ_{null} (here = 0), shown in different colors; the probability of direction (pd) is the larger of the two posterior probabilities.



Because *pd* is derived from the posterior, it inherits dependence on the full model specification (likelihood and prior), so the long-run error behavior of *pd*-based decisions is not a property of *pd* "in isolation" but of the overall Bayesian procedure that generates the posterior.

Probability of Direction as a test of existence and direction

A convenient starting point for calibration is the mapping between pd and the frequentist p-value. Under flat (or sufficiently diffuse symmetric) priors in typical hypothesis-testing scenarios (e.g., testing means), pd is the posterior probability of the data-favored sign relative to θ_{null}. This value corresponds to the complement of the one-sided p-value (Casella & Berger, 1987; Marsman & Wagenmakers, 2017), with the two-sided relation approximated by $p \approx 2p = 2(1 - pd)$. The pd is therefore often used to summarize both the direction and existence of an effect [**REFs**].

This interpretation is clearest when phrased as a decision rule $pd > c$: the posterior must assign more than c probability mass to the side of θ_{null} favored by the data. Setting $c = 1 - \alpha/2$ is convenient because it bounds the posterior mass in the opposite tail by $\alpha/2$. For concreteness, suppose the posterior favors a positive effect so that $pd = \Pr(\theta > \theta_{\text{null}} \mid y)$. Then $pd > c$ implies

$$\Pr(\theta \leq \theta_{\text{null}} \mid y) = 1 - \Pr(\theta > \theta_{\text{null}} \mid y) < 1 - c = \alpha/2.$$

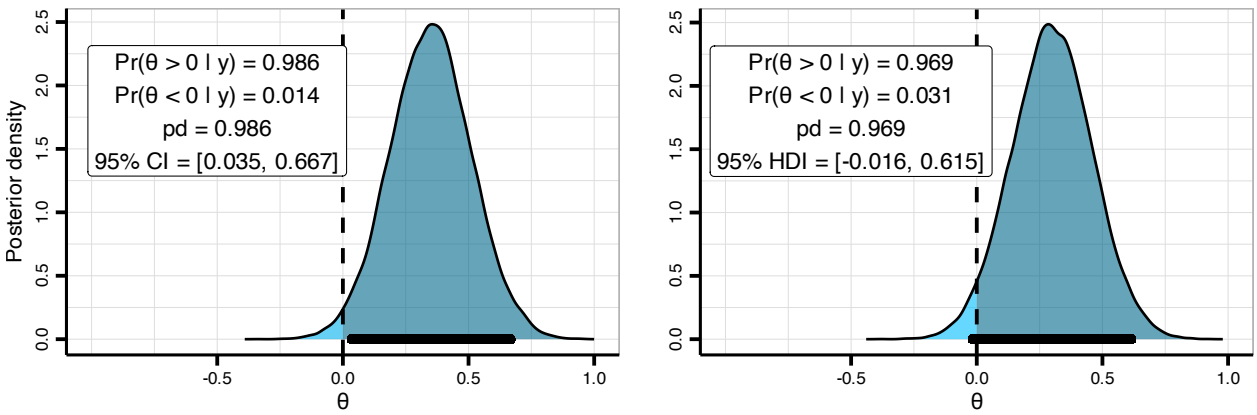
With $\alpha = 0.05$, this gives $c = 0.975$: the rule $pd > 0.975$ requires that $\Pr(\theta \leq \theta_{\text{null}} \mid y) < 0.025$, placing less than 2.5% of posterior mass at or below θ_{null}.

Let q_p denote the p-th posterior quantile, so the central $(1 - \alpha)$ equal-tailed credible interval is $[q_{\alpha/2}, q_{1-\alpha/2}]$, which leaves probability $\alpha/2$ in each tail. The condition $\Pr(\theta \leq \theta_{\text{null}} \mid y) < \alpha/2$ holds if and only if θ_{null} falls below $q_{\alpha/2}$, that is, when $\theta_{\text{null}} \notin [q_{\alpha/2}, q_{1-\alpha/2}]$. Thus, observing $pd > 0.975$ is equivalent to excluding θ_{null} from the central 95% equal-tailed credible interval (see Figure 2).

Given these considerations, although pd and the frequentist p-value target different quantities, in the case considered here it is plausible that decision rules of the form $pd > c$ approximate the operating characteristics of familiar frequentist rules such as $p < \alpha$.

Figure 2

Posterior distributions illustrating the relationship between the probability of direction (pd) and 95% Credible Interval (CI), using $c = 0.975$ and $\theta_{null} = 0$.



Simulation study

In our simulations we generate data under the global null (all $\theta = 0$) and we compute the frequency of $pd > c$, where $c = 0.975$. Operationally, this quantifies how often the procedure falsely excludes zero from the 95% central credible interval (bounded by $(1 - c)$ and c), that is, incorrectly concludes that an effect exists (with a particular sign).

Although *pd* is a measure of direction, pure directional testing conditions on the assumption that $\theta \neq 0$ and evaluates only the sign of the effect by comparing H_+ versus H_-; $H_0 : \theta = 0$ is not among the hypotheses under consideration. In contrast, here we explicitly include H_0 (Goeman et al., 2010), focusing on decisions about both existence and direction (when the interval excludes zero, e.g., H_+) rather than decisions about direction only (when the interval includes zero, e.g., H_{0+}).

Specifically, we conduct a Monte Carlo simulation in which, for each condition, we generated families of size $m \in \{1, 2, 5, 10, 20\}$, with $n \in \{30, 50, 100\}$ observations per dataset, and repeated the procedure 10,000 times. In every replication, data were generated under the global null: each dataset had true mean 0 and unit variance, $y \sim \mathcal{N}(0, 1)$. For each dataset, we compute the posterior distribution of θ assuming $y_i \mid \theta \sim \mathcal{N}(\theta, 1)$ and $\theta \sim \mathcal{N}(0, \tau^2)$, with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$ ranging from skeptical to relatively diffuse priors.

In most applications, the *pd* is estimated from posterior draws as the larger of the two one-sided posterior probabilities: the proportion of draws above θ_{null} and the proportion below θ_{null}. Because this is a Monte Carlo estimate, its numerical resolution depends directly on the number of posterior draws (Makowski et al., 2019).

In our simulation, however, the conjugate normal–normal model implies a Gaussian posterior, $\theta \mid y \sim \mathcal{N}(\mu_{\text{post}}, \sigma^2_{\text{post}})$, allowing *pd* to be computed in closed form. Specifically,

$$pd = \max(\Pr(\theta > \theta_{null} \mid y), \Pr(\theta < \theta_{null} \mid y)) = \Phi\left(\left|\frac{\mu_{\text{post}} - \theta_{null}}{\sigma_{\text{post}}}\right|\right),$$

with $\theta_{\text{null}} = 0$.

For the frequentist comparison, we conducted two-sided one-sample z-tests.

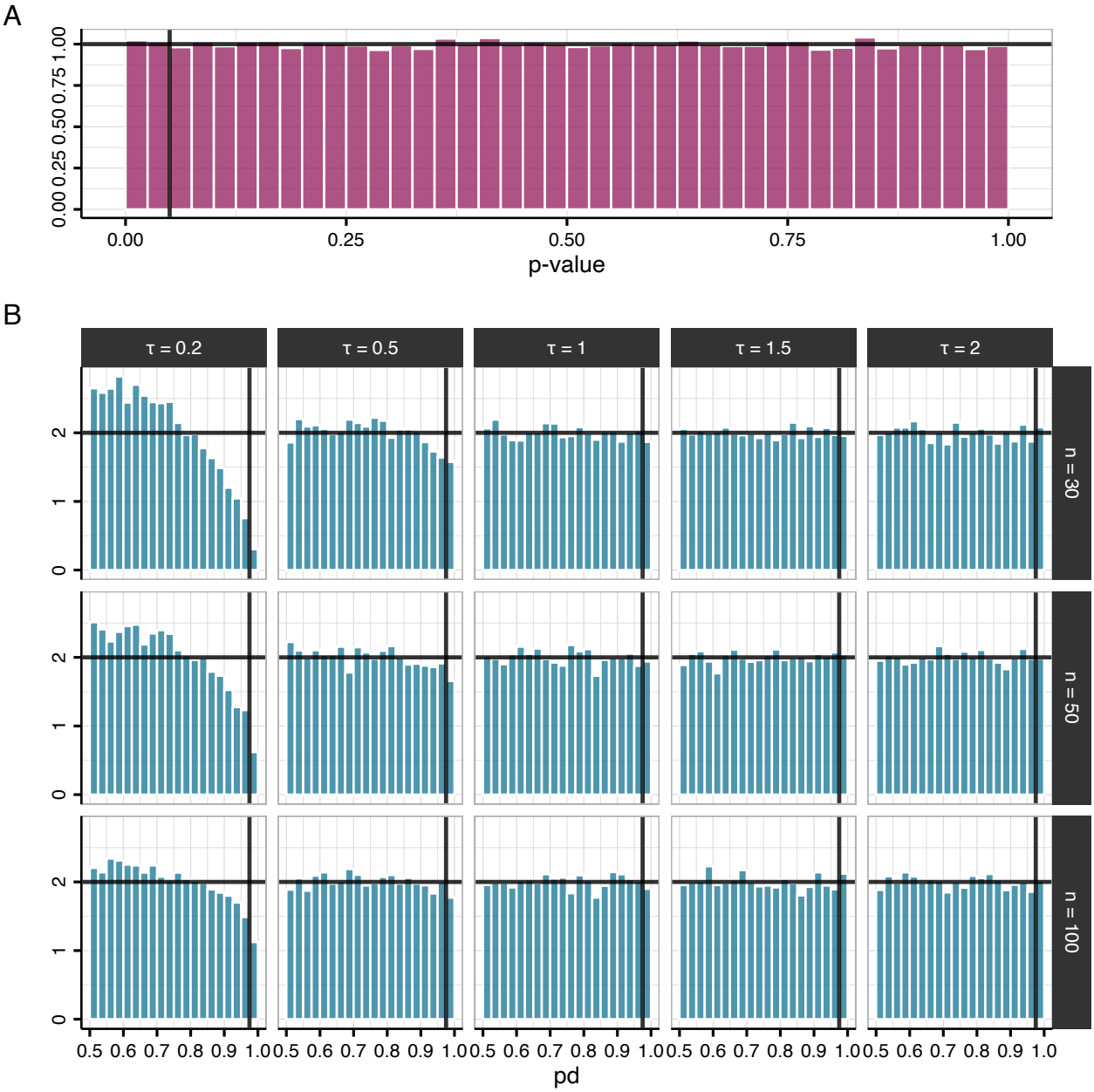
Prior influence

Under the null hypothesis, p-values are uniformly distributed on $[0, 1]$, and because *pd* can be mapped to the p-value, this implies a corresponding null reference behavior for *pd* on $[0.5, 1]$. However, because the *pd* is a posterior probability about the sign of θ, its null distribution is not determined by the likelihood alone: it can shift depending on how strongly the prior concentrates probability near 0 (i.e., how readily the prior "expects" near-null effects). Showing this prior dependence at $m = 1$ makes the target of later multiplicity analyses explicit: even before any multiplicity is introduced, a fixed cutoff such as $pd > 0.975$ does not have a universal "null meaning" across priors and sample sizes.

As shown in Figure 3, *pd* exhibits behavior broadly analogous to a p-value when samples are large and the prior places relatively little mass near 0. However, increasing prior skepticism (i.e., concentrating more prior mass around 0) reduces the frequency of extreme *pd* values.

Figure 3

Null reference distributions for p-values and probability of direction. Panel A shows the histogram of two-sided one-sample z-test p-values from 10,000 replications under the global null ($\theta = 0$), with binwidth 0.025. Panel B shows the corresponding empirical distributions of the probability of direction (pd) computed from a conjugate normal posterior, stratified by sample size ($n = 30, 50, 100$; rows) and prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$; columns). The vertical dashed line marks the commonly used evidential cutoff (pd = 0.975), which corresponds to a two-sided p-value of 0.05 under the standard conversion $p = 2(1 - pd)$.



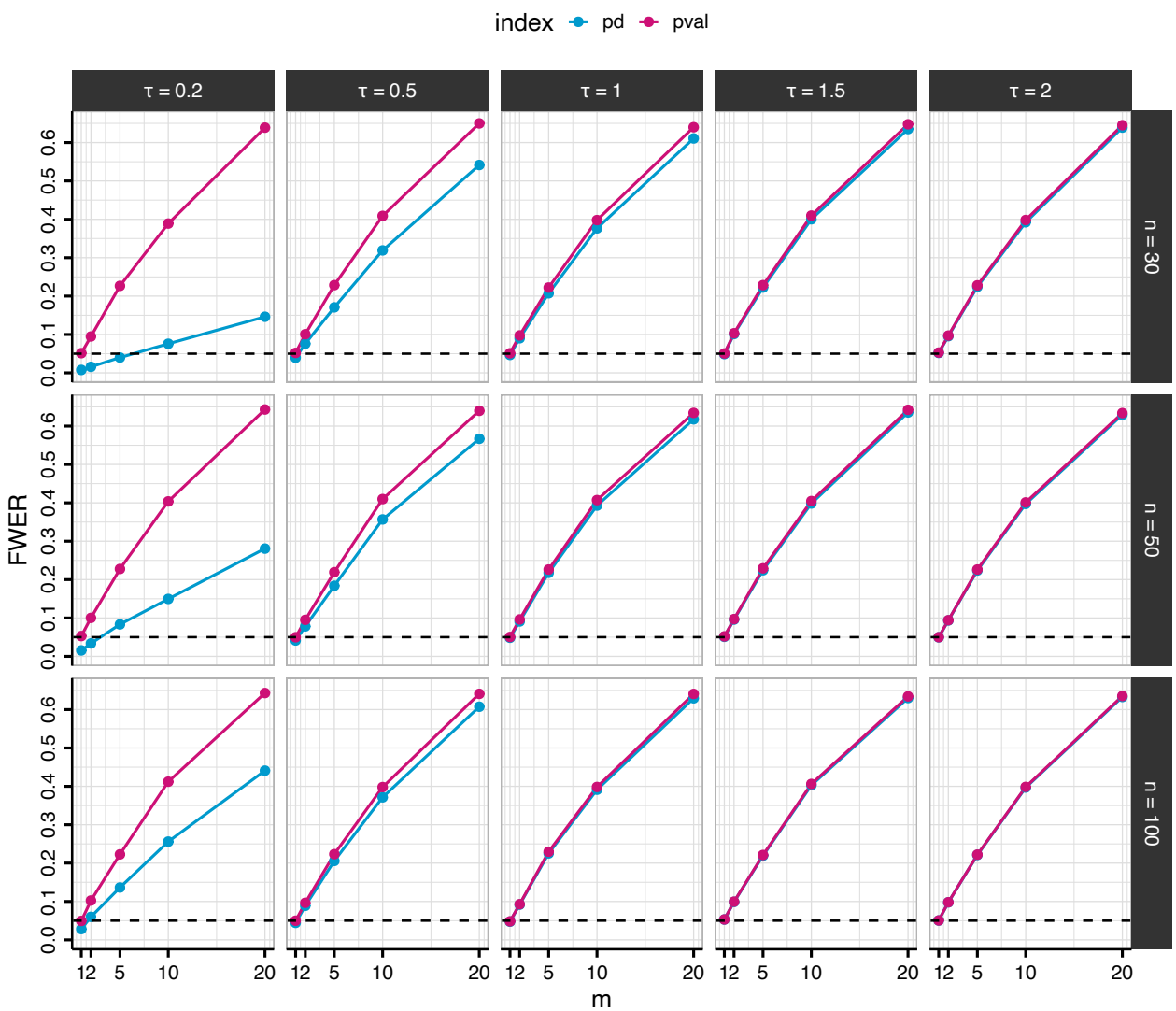
Behavior under multiplicity

Building on this baseline, multiplicity can be introduced as a second, distinct mechanism: even when each individual test has only a small probability of producing an extreme pd under H_0, the probability that at least one among $pd_1, \ldots, pd_m$ is extreme increases with the number of tests m. We quantify this familywise error rate (FWER) behavior by estimating the frequency of $pd > 0.975$ for families ranging from $m = 1$ to $m = 20$.

As expected, pd-based decisions behaves broadly analogously to their frequentist counterpart when sample sizes are large and the prior places relatively little mass near 0 (Figure 4). By contrast, as the prior becomes more skeptical (i.e., increasingly concentrated around 0), the FWER decreases.

Figure 4

Familywise error rates under the global null for p-values and probability of direction. The figure shows the estimated familywise error rate (FWER), defined as the probability of at least one false positive across m tests, as a function of the number of tests (m), based on 10,000 replications under the global null ($\theta = 0$ for all tests). The pink curve corresponds to declaring at least one discovery when at least one two-sided one-sample z-test p-value < 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd > 0.975 under the conjugate normal Bayesian model. Rows vary the per-test sample size ($n = 30, 50, 100$), and columns vary the prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$).



The simulation results suggest that Type I error rates could be reduced by adopting more skeptical within-model priors for each parameter. However, using within-model prior skepticism as a general remedy for multiplicity is problematic because it conflates two distinct goals: (i) encoding substantive prior knowledge about a specific parameter and (ii) calibrating the error rate of a decision rule under repeated testing.

When the primary aim is error-rate control, adjusting the within-model prior introduces bias into parameter estimates as an unintended side effect. The approach developed below instead adjusts the prior odds assigned to the multiple hypotheses while keeping the within-model prior for any single parameter unchanged. This separation allows researchers to choose the most appropriate prior for each parameter based on substantive knowledge and modeling considerations, while still controlling error rates when making simultaneous statements about multiple parameters.

Importantly, the method proposed below is valid if and only if the within-model prior is centered symmetrically around the tested null value. That is, the prior must allow the parameter to take values both greater than and smaller than the null value with equal prior plausibility, rather than constraining the parameter space to only the expected direction. Subject to this symmetry requirement, researchers retain full flexibility in prior specification.

Probability of direction, posterior odds, and prior-odds adjustment

From posterior probabilities to posterior odds

To connect pd to posterior odds, consider the directional hypotheses $H_+ : \theta > 0$ and $H_- : \theta < 0$. These hypotheses partition the parameter space based on the sign of θ. If the prior for θ is continuous and symmetric about 0, then the two directions are equally plausible a priori: $\Pr(H_+) = \Pr(H_-) = 1/2$. Without loss of generality, assume the posterior favors H_+, so that $pd = \Pr(H_+ \mid y) = \Pr(\theta > 0 \mid y)$, then the complement is given by $1 - pd = \Pr(\theta \leq 0 \mid y)$. The posterior odds comparing these two hypothesis are therefore:

$$\frac{pd}{1 - pd} = \frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)}.$$

For a continuous posterior distribution, the probability of any single point (e.g., $\Pr(\theta = 0 \mid y)$) is 0, which means that one could write $\frac{pd}{1-pd} = \frac{\Pr(H_+|y)}{\Pr(H_-|y)}$. However, we retain the notation H_{-0} in the denominator to explicitly indicate that the hypothesis includes the point null value of zero.

Applying Bayes' rule, these posterior odds can be decomposed into a Bayes Factor and prior odds:

$$\frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)} = \mathrm{BF}_{+/-0} \times \frac{\Pr(H_+)}{\Pr(H_{-0})},$$

where BF+/−0 quantifies the relative likelihood of the data under $H+$ versus H_{-0}. Under prior symmetry, $\frac{\Pr(H_+)}{\Pr(H_{-0})} \approx 1$, which yields $\frac{pd}{1-pd} = \mathrm{BF}_{+/-0}$ (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Thus, the odds $\frac{pd}{(1-pd)}$ computed from posterior draws can be interpreted directly as evidence favoring $\theta > 0$ over $\theta \leq 0$ (or $\theta < 0$ over $\theta \geq 0$).

Multiplicity adjustment via post-hoc prior-odds

Assessing a single directional claim under a symmetric prior centered at θ_{null} corresponds to prior indifference (i.e., $1/2$). If m directional claims are considered simultaneously and treated as independent a priori, assigning probability $1/2$ to each claim induces a joint prior that quickly assigns high probability to the event that at least one effect is in the claimed direction. Specifically, the prior probability that none of the m effects is in the claimed direction is $(1/2)^m$, so the prior probability that at least one is in the claimed direction is $1 - (1/2)^m$. For example, with $m = 10$, Pr(none in claimed direction) $= 0.5^{10} \approx 0.001$, and Pr(at least one in claimed direaction) $= 1 - 0.5^{10} \approx 0.999$.

In this setting, a principled response to multiplicity is to reduce the prior odds assigned to any particular directional claim as a function of m. Because the prior is symmetric in the sign of θ and pd is defined as the posterior probability of the dominant direction, the same prior-odds adjustment applies whether the data favor $\theta > 0$ or $\theta < 0$.

Concretely, for each test $i = 1, \ldots, m$, let $\Pr(H_{1_i})$ denote the multiplicity-adjusted prior probability of the favored direction and let $\Pr(H_{0_i})$ denote the prior probability of its complement (i.e., the opposite direction, with 0 included), with $\Pr(H_{0_i}) + \Pr(H_{1_i}) = 1$. Then the adjusted

posterior odds satisfy

$$\frac{pd_{\text{adj},i}}{1 - pd_{\text{adj},i}} = \frac{pd_i}{1 - pd_i} \times \frac{\Pr(H_{1_i})}{\Pr(H_{0_i})},$$

or equivalently,

$$pd_{\text{adj},i} = \frac{pd_i \ \Pr(H_{1_i})}{pd_i \ \Pr(H_{1_i}) + (1 - pd_i) \ \Pr(H_{0_i})}.$$

Thus, multiplicity acts only through the prior-odds factor $\Pr(H_{1_i})/\Pr(H_{0_i})$.

Choosing the prior-odds

The multiplicity adjustment above operates entirely through the prior-odds factor $\Pr(H_{1,i})/\Pr(H_{0,i})$. To specify these prior odds in a way that scales with the number of tests, we follow the global-null formulation of Jeffreys (1938) (see also Westfall et al. (1997)).

Let Q denote the global null event that none of the m parameters lies in its data-favored direction, and let Q^c denote its complement (i.e., at least one parameter lies in its data-favored direction). We assign equal prior probability to these events, $\Pr(Q) = \Pr(Q^c) = 0.5$.

For each test $i = 1, \ldots, m$, define $H_{1,i}$ as the directional claim corresponding to the posterior-dominant sign (i.e., the direction used in $pd_i = \Pr(H_{1,i} \mid y)$), and let $H_{0,i}$ denote its complement (the opposite direction, with 0 included). With these definitions,

$$Q = \bigcap_{i=1}^{m} H_{0,i}.$$

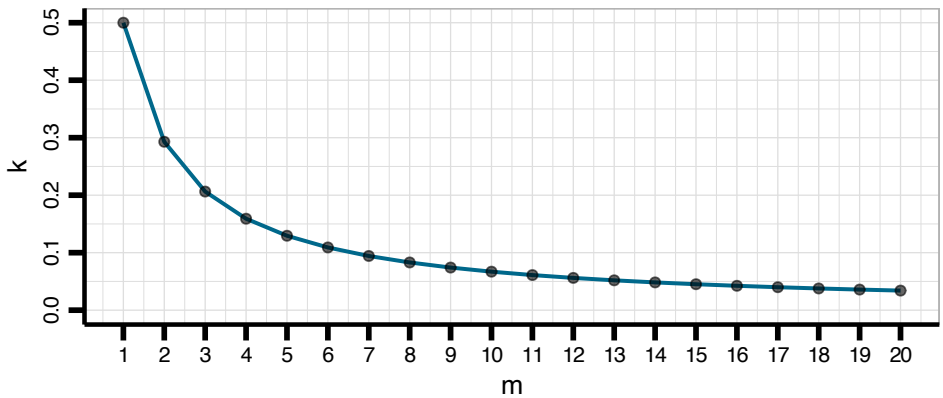
Assuming prior independence across tests and a common per-hypothesis prior probability $\Pr(H_{1,i}) = k$, it follows that $\Pr(Q) = (1 - k)^m$. Imposing $\Pr(Q) = 0.5$ yields $(1 - k)^m = 0.5$, and therefore

$$\Pr(H_{1_i}) = k \ = \ 1 - 0.5^{1/m}.$$

As m increases, $k = \Pr(H_{1,i})$ decreases, encoding increasing skepticism about any particular directional claim (see Figure 5).

Figure 5

Prior probabilty k for each $H_{1,i}$ (where $i = 1, \ldots, m$) as a function of the total number of hypotheses m, assuming the joint null hypothesis has a prior probability of 0.5.

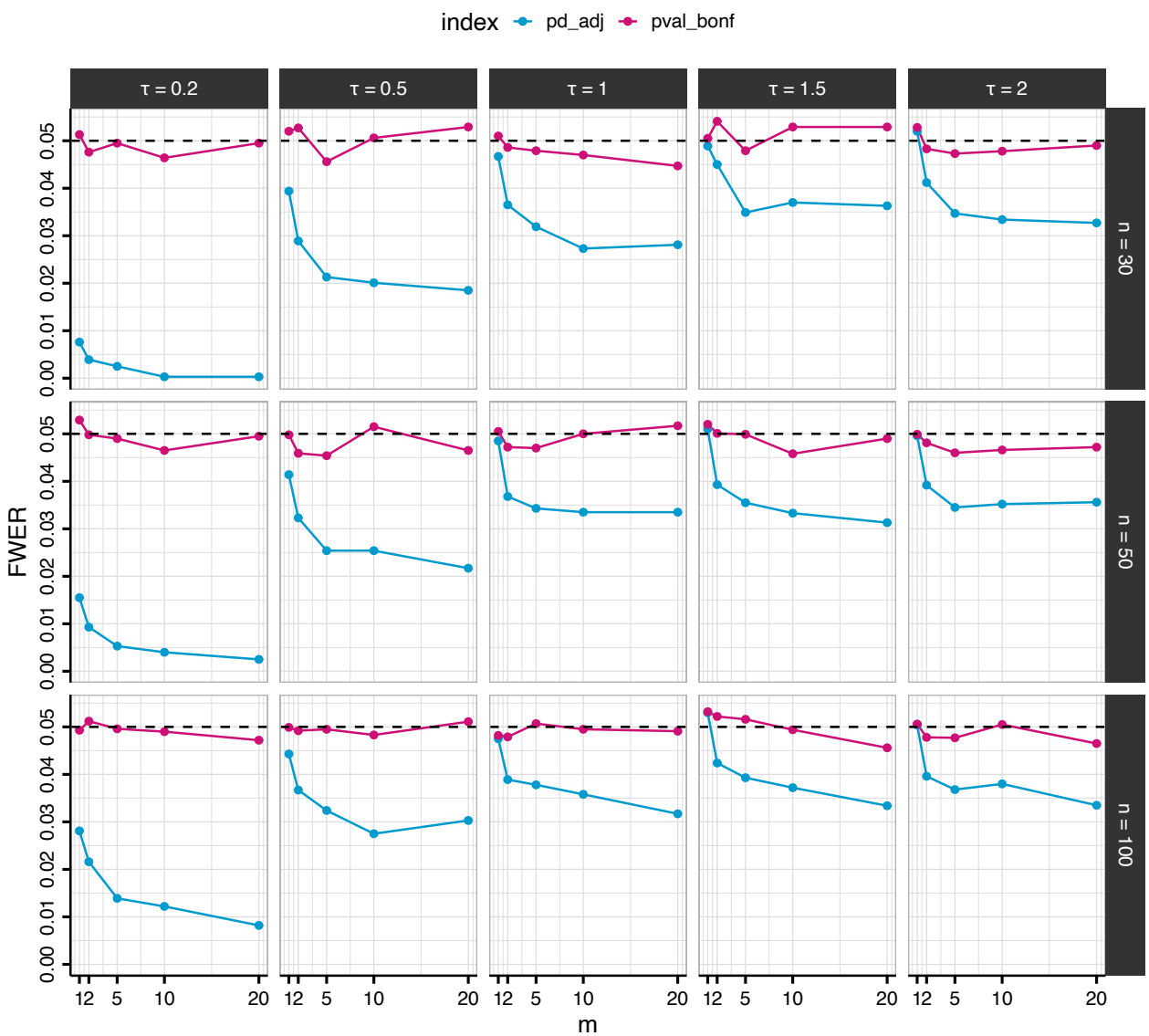


Following Westfall et al. (1997), we compare this method with the Bonferroni p-value adjustment ($p_{adj} = p \times m$), which gives strong frequentist family-wise error rate control.

Although grounded in Bayesian prior-odds logic, the procedure exhibits behavior similar to frequentist multiplicity correction. However, it tends to be more conservative, even when both n and τ are large (Figure 6). Importantly, when skeptical priors are already used for estimation, applying the multiplicity adjustment constitutes a double correction.

Figure 6

Familywise error rates under the global null for adjusted p-values and probability of direction. The figure shows the estimated familywise error rate (FWER), defined as the probability of at least one false positive across m tests, as a function of the number of tests (m), based on 10,000 replications under the global null ($\theta = 0$ for all tests). The red curve corresponds to declaring at least one discovery when at least one two-sided one-sample z-test p-value < 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd > 0.975 under the conjugate normal Bayesian model. Rows vary the per-test sample size ($n = 30, 50, 100$), and columns vary the prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$).



Having linked the multiplicity adjustment to the global null event Q, the same construction yields a more general calibration directly in terms of $\Pr(Q)$. Specifically, instead of fixing $\Pr(Q) = 0.5$, researchers may encode background knowledge by choosing $\Pr(Q) = q$. As stated above, under a priori independence and a common per-hypothesis prior probability $k = \Pr(H_{1,i})$ for the favored-direction alternative, $\Pr(Q) = (1-k)^m$, so choosing q implies

$$(1-k)^m = q \quad \Rightarrow \quad k = 1 - q^{1/m} = \Pr(H_{1,i}).$$

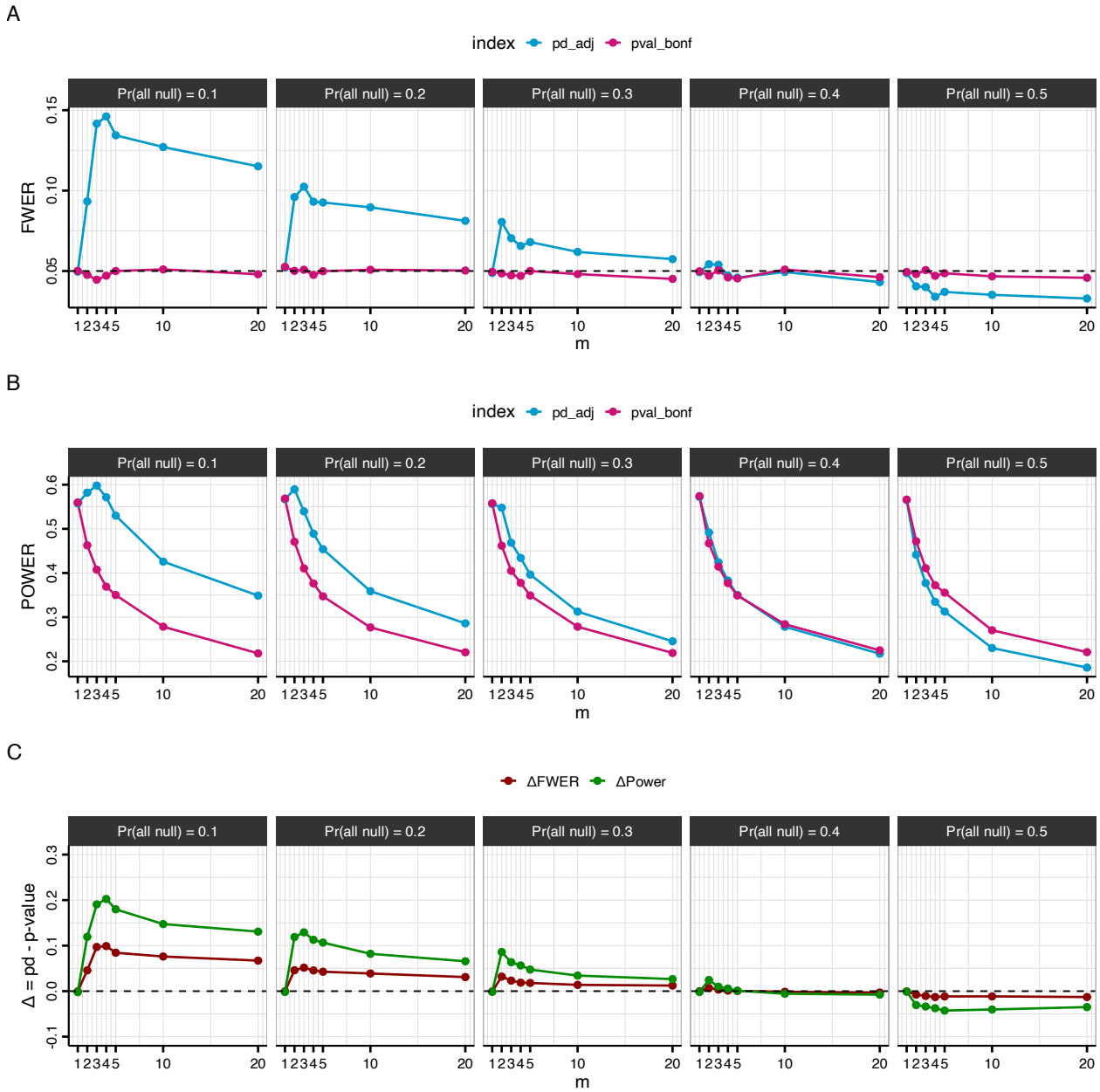
Prior knowledge guides q: when few (if any) non-null effects are expected, q should be large; when effects are well established, q can be smaller. For small m and sufficiently small q, however, the calibration can become anti-conservative: $\Pr(H_{0,i}) = 1 - k = q^{1/m}$ may fall below $1/2$, which implies $\Pr(H_{1,i}) > 1/2$ and therefore increases rather than decreases the odds for $H_{1,i}$. Because the goal of the adjustment is to guard against inflated false positives, it should never move posterior probabilities in this direction; accordingly, a simple safeguard is to leave posterior probabilities unchanged whenever $\Pr(H_{0,i}) < 1/2$ (equivalently, whenever $q^{1/m} < 1/2$).

Once q is fixed, the prior odds assigned to each individual hypothesis become a deterministic function of the family size m: larger families imply smaller k, and consequently a higher prior probability on the null for each test.

To illustrate the impact of q on both FWER and power, we consider two data-generating regimes: a global-null scenario in which $\theta_1 = \cdots = \theta_m = 0$, and an alternative scenario in which all but one effect are null ($\theta_1 = \cdots = \theta_{m-1} = 0$, $\theta_m = 0.3$). We then repeat the previous simulation procedure with $\tau = 2$ and $n = 50$, i.e. a setting in which *pd* and *p*-value behavior is broadly comparable, while varying only the number of tests $m \in \{1, 2, 3, 4, 5, 10, 20\}$ and the prior parameter $q \in \{0.50, 0.40, 0.30, 0.20, 0.10\}$.

Figure 7

Familywise error rate (Panel A), power (Panel B), and difference between method (Panel C) for adjusted p*-values and probability of direction as a function of the number of tests (m) and prior* $\Pr(\textit{all } \theta = 0)$. *The Bonferroni-adjusted* p*-value curve (pink) declares at least one discovery when any two-sided one-sample* z*-test satisfies* $p < 0.05$*, whereas the adjusted* pd *curve (blue) declares at least one discovery when any* $pd > 0.975$ *under the conjugate normal Bayesian model. Panel C shows the difference (adjusted pd minus Bonferroni-adjusted p-value) for FWER (under the global null) and power (under the global alternative).*



While Bonferroni-adjusted p-values strictly control the FWER at the nominal level (Figure 7, Panel A), this rigor comes at the cost of substantially diminished power (Figure 7, Panel B). In contrast, the adjusted *pd* procedure offers a flexible trade-off governed by the prior parameter q. For lower values ($q \in \{0.10, 0.20, 0.30\}$), the exchange is favorable: the resulting gain in power outpaces the marginal increase in FWER (Figure 7, Panel C). Notably, setting $q = 0.4$ produces FWER and power trajectories that mirror the Bonferroni adjustment. This empirical alignment is consistent with Westfall et al. (1997) asymptotic proof that a global null probability of $q = e^{-1} \approx 0.368$ reproduces Bonferroni-like behavior. Thus, selecting $q \approx 0.4$ effectively calibrates the Bayesian prior to match the severity of the frequentist α correction.

Accounting for correlation

A limitation of the proposed procedure is that it allocates the global-null probability across m hypotheses under an independence assumption. When hypotheses are correlated, for example, because parameters are estimated jointly in a single model or because measures share the same observations, independence-based multiplicity adjustments can be overly conservative (Pesarin & Salmaso, 2010; Westfall & Young, 1993).

A pragmatic solution is to replace the nominal number of hypotheses m with an effective number m_{eff} that reflects the actual amount of unique information (Cheverud, 2001; Li & Ji, 2005). This approach requires estimating a correlation matrix $\mathbf{R}$ that captures the dependencies among posterior distributions, then mapping $\mathbf{R}$ to m_{eff} using an eigenvalue-based method. Multiple methods exist for estimating m_{eff}, each with different assumptions. Here we apply the eigenvalue-based approach of Cheverud (2001) (see Appendix X for computational details and Del Giudice (2021) for critiques).

To evaluate whether this adjustment reduces conservativeness under dependence while maintaining adequate familywise error rate (FWER) control, we extended our simulation study to correlated data. We conducted a Monte Carlo simulation generating families of size $m \in \{2, 4, 6, 10, 20\}$ with $n = 50$ observations per dataset across 10,000 replications per condition.

In each replication, true parameter values were drawn from a multivariate normal

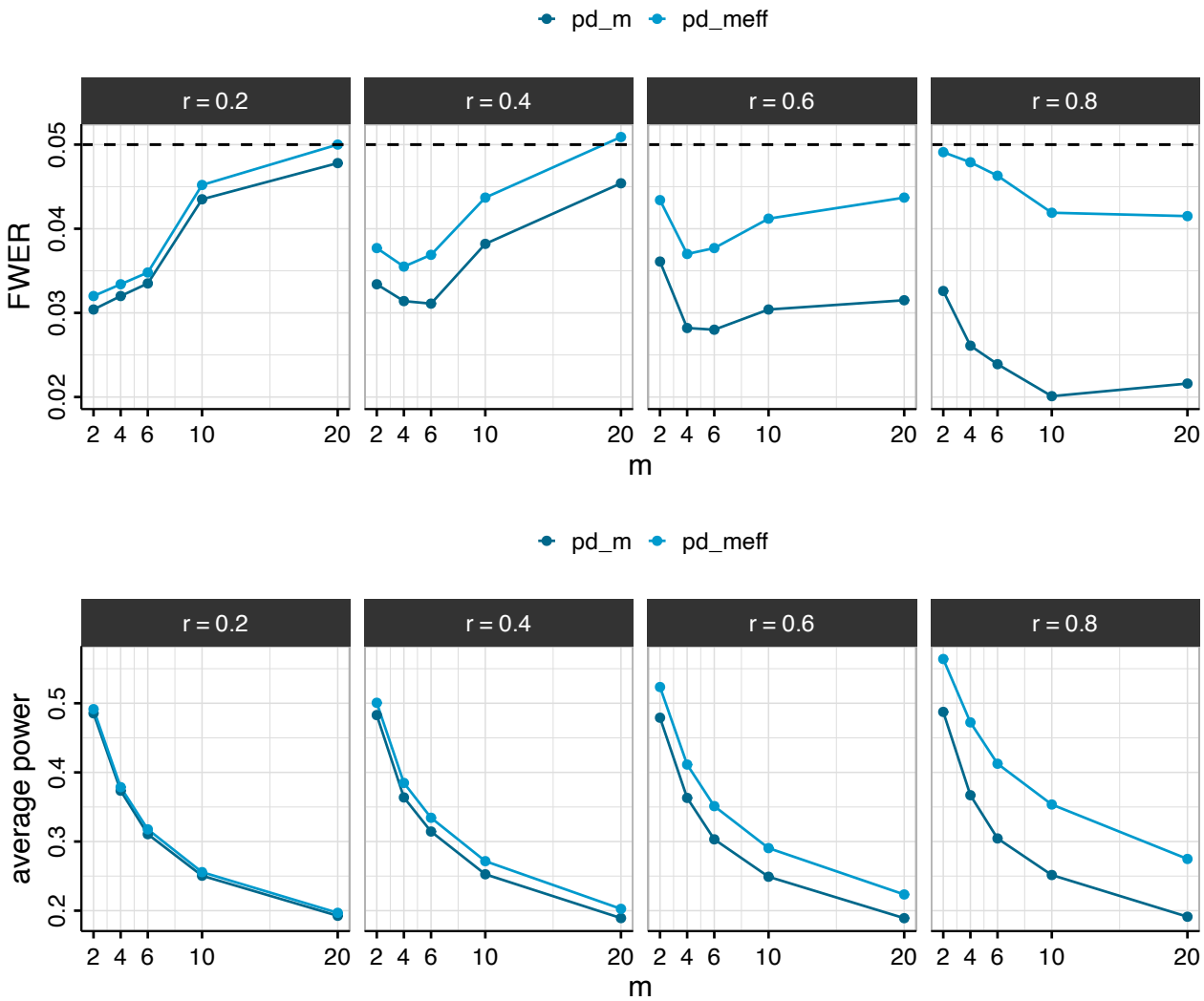
distribution with equicorrelation structure: $\theta \sim \mathcal{N}(\mu, \Sigma)$, where $\mu_1 = \cdots = \mu_{m/2} = 0$ (null hypotheses) and $\mu_{m/2+1} = \cdots = \mu_m = 0.3$ (non-null hypotheses). The correlation matrix $\mathbf{R}$ had constant pairwise correlation $\rho \in \{0.2, 0.4, 0.6, 0.8\}$.

For each dataset, we obtained posterior distributions using a multivariate normal conjugate model with $\tau = 2$, and for the adjustment we set the prior parameter $q = 0.4$ based on previous results. We computed FWER under the null subset and average power (Benjamini & Hochberg, 1995; Izmirlian et al., 2025), the proportion of true discoveries under $\mu = 0.3$. We compared the multiplicity adjustment procedure when accounting for correlations (using m_{eff}) versus ignoring them (using m).

As shown in Figure 8, the method accounting for correlation provided better performance, particularly under high correlation scenarios, by maintaining proper FWER control while preserving greater power. Without this adjustment, the procedure became excessively conservative as correlation increased, controlling FWER well below the nominal level but at substantial cost to power.

Figure 8

Familywise error rate (top panels) and average power (bottom panels) for two adjusted probability of direction methods (pd_m and pd_meff) as a function of the number of tests (m) and correlation structure (r). pd_meff incorporates an effective number of tests correction to account for correlation structureEach column represents a different correlation level.



Application

…

Discussion

Although *pd* is a descriptive posterior index, thresholding it into a decision rule and applying that rule repeatedly across many parameters induces a multiplicity problem, as the

probability of at least one false claim increases with the number of decisions. The multiplicity adjustment proposed here targets this decision-level error while allowing researcher-specific knowledge to enter transparently through the prior-odds specification.

A key practical advantage of this procedure is its applicability to *pd* values from different modeling strategies. While the most coherent approach analyzes all relevant data within a single joint model that explicitly represents dependence (Gelman et al., 2012, 2013; Kruschke, 2011), this is not always feasible. Researchers often report *pd* values from multiple separate analyses within the same manuscript [**REFs**]. The procedure accommodates both settings: when a joint model is available, its posterior correlation structure can be leveraged; when only separate analyses are available, the adjustment proceeds under an independence assumption.

Additionally, the presented procedure can be generalized to cases with non-zero null values (Lakens et al., 2018). The generalization is achieved by defining directional hypotheses relative to any chosen value, provided that within-model priors for the tested parameters are set symmetrically around this value. While point-null hypotheses (testing whether a parameter equals exactly one value) have been criticized as often unrealistic [**ADD REFs**], and interval-based approaches offer alternative frameworks for hypothesis evaluation, those considerations extend beyond our current focus. We acknowledge that various methods exist for Bayesian hypothesis testing, each with distinct advantages and limitations.

An important caveat is that this framework does not provide evidence for a point null hypothesis (i.e., posterior support for θ_{null}); instead, it evaluates directional claims based on inequalities (e.g., $\theta > \theta_{null}$ vs. $\theta \leq \theta_{null}$). By construction, the complementary hypothesis includes the null value, but the framework does not assign it special status. When the null value falls outside the credible interval, one can infer its relative implausibility as a candidate value, but this differs fundamentally from quantifying evidence in favor of the null. This contrasts with Bayes factor approaches that treat the point null as a distinct model by assigning it positive prior mass, thereby enabling direct statements of evidence supporting the null hypothesis [Heck et al. (2023); Morey and Rouder (2011); Ly et al. (2016); **add dienes**].

Lastly, one might argue that when the focus is on estimation rather than hypothesis testing, multiplicity adjustment is unnecessary. However, reported results inevitably influence future studies (both our own and others'), guiding which findings warrant replication, which theories merit further development, and which interventions should be implemented. This influence constitutes an implicit form of decision-making. Our argument is not that researchers must always frame their work as explicit hypothesis testing, but that they should consider the consequences of ignoring multiplicity and be transparent about the implied prior probabilities assigned to each hypothesis when making decisions across multiple tests.

For applied reporting, a two-tier strategy is recommended: (i) report the unadjusted *pd* for each parameter, and (ii) report at least one multiplicity-adjusted measure based on the proposed prior-odds correction. When *pd* values are computed from posterior distributions obtained from different models, although just as an approximation a conservative default is to report the adjustment induced by $\Pr(Q) = 0.4$, a very skeptical but defensible choice that yields corrections analogous to the Bonferroni adjustment for p-values. When substantive knowledge motivates alternative priors, report both the conservative and the informed adjustment to demonstrate robustness to plausible prior assumptions.

Despite its limitations, our method provides a coherent, practical way to recalibrate *pd*-based decisions for multiple comparisons, making reported claims less likely to reflect chance patterns across many parameters.

Appendix X: Eigenvalue-based approach

To account for dependence among hypotheses, we follow the eigenvalue-based approach of Cheverud (2001). This method adjusts the nominal number of tests m to an effective number m_{eff} that reflects the degree of correlation among test statistics.

Estimation procedure

1. **Obtain the correlation matrix**: From the posterior covariance matrix $\mathbf{\Sigma}$ of the multivariate normal approximation to the joint posterior of the m parameters of interest, we derive the correlation matrix $\mathbf{R}$.

2. **Extract eigenvalues**: We compute the eigenvalues $\lambda_1, \ldots, \lambda_m$ of $\mathbf{R}$.

3. **Calculate eigenvalue variance**: We compute the observed variance of the eigenvalues as:

$$V\lambda_{\text{obs}} = \frac{\sum_{i=1}^{m} (\lambda_i - 1)^2}{m - 1}$$

4. **Estimate effective number of tests**: Following Cheverud (2001), the effective number of independent tests is:

$$m_{\text{eff}} = m \left(1 - \frac{(m-1)V\lambda_{\text{obs}}}{m^2} \right)$$

This formula quantifies the proportional reduction in the number of tests based on the ratio of eigenvalue variance to its maximum. The resulting m_{eff} ranges between 1 (perfect correlation among all tests) and m (complete independence), with intermediate values reflecting partial dependence.

References

Albers, C. J., Kiers, H. A., & Ravenzwaaij, D. van. (2018). Credible confidence: A pragmatic view on the frequentist vs Bayesian debate. *Collabra: Psychology*, *4*(1).

Altoè, G., Bertoldo, G., Zandonella Callegher, C., Toffalini, E., Calcagnì, A., Finos, L., & Pastore, M. (2020). Enhancing statistical inference in psychological research via prospective and retrospective design analysis. *Frontiers in Psychology*, *Volume 10 - 2019*. https://doi.org/10.3389/fpsyg.2019.02893

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, *57*(1), 289–300. https://doi.org/https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Berger, J. O., & Sellke, T. (1987). Testing a point null hypothesis: The irreconcilability of p values and evidence. *Journal of the American Statistical Association*, *82*(397), 112–122.

Bernardo, J. M., & Smith, A. F. M. (1994). *Bayesian Theory* (1st ed.). Wiley. https://doi.org/10.1002/9780470316870

Berry, D. A., & Hochberg, Y. (1999). Bayesian perspectives on multiple comparisons. *Journal of Statistical Planning and Inference*, *82*(1), 215–227. https://doi.org/10.1016/S0378-3758(99)00044-0

Box, G. E. P., & Tiao, G. C. (1992). *Bayesian inference in statistical analysis* (1st ed.). Wiley. https://doi.org/10.1002/9781118033197

Casella, G., & Berger, R. L. (1987). Reconciling bayesian and frequentist evidence in the one-sided testing problem. *Journal of the American Statistical Association*, *82*(397), 106–111. https://doi.org/10.1080/01621459.1987.10478396

Chen, G., Taylor, P. A., Cox, R. W., & Pessoa, L. (2020). Fighting or embracing multiplicity in neuroimaging? Neighborhood leverage versus global calibration. *NeuroImage*, *206*, 116320. https://doi.org/10.1016/j.neuroimage.2019.116320

Cheverud, J. M. (2001). A simple correction for multiple comparisons in interval mapping

genome scans. *Heredity*, *87*(1), 52–58. https://doi.org/https://doi.org/10.1046/j.1365-2540.2001.00901.x

Cohen, J. (1994). The earth is round (p < .05). *American Psychologist*, *49*(12), 997–1003. https://doi.org/10.1037/0003-066X.49.12.997

Dawid, A. P. (1994). Selection paradoxes of bayesian inference. *Lecture Notes-Monograph Series*, *24*, 211–220.

Del Giudice, M. (2021). Effective dimensionality: A tutorial. *Multivariate Behavioral Research*, *56*(3), 527–542.

Edwards, W., Lindman, H., & Savage, L. J. (1963). Bayesian statistical inference for psychological research. *Psychological Review*, *70*(3), 193.

Erčulj, F., & Štrumbelj, E. (2015). Basketball shot types and shot success in different levels of competitive basketball. *PLOS ONE*, *10*(6), 1–14. https://doi.org/10.1371/journal.pone.0128885

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian data analysis* (3rd ed.). Chapman; Hall/CRC. https://doi.org/10.1201/b16018

Gelman, A., Hill, J., & Yajima, M. (2012). Why we (usually) don't have to worry about multiple comparisons. *Journal of Research on Educational Effectiveness*, *5*(2), 189–211. https://doi.org/10.1080/19345747.2011.618213

Gelman, A., & Tuerlinckx, F. (2000). Type s error rates for classical and bayesian single and multiple comparison procedures. *Computational Statistics*, *15*(3), 373–390. https://doi.org/10.1007/s001800000040

Gigerenzer, G. (2004). Mindless statistics. *The Journal of Socio-Economics*, *33*(5), 587–606. https://doi.org/https://doi.org/10.1016/j.socec.2004.09.033

Goeman, J. J., Solari, A., & Stijnen, T. (2010). Three-sided hypothesis testing: Simultaneous testing of superiority, equivalence and inferiority. *Statistics in Medicine*, *29*(20), 2117–2125. https://doi.org/https://doi.org/10.1002/sim.4002

Heck, D. W., Boehm, U., Böing-Messing, F., Bürkner, P.-C., Derks, K., Dienes, Z., Fu, Q., Gu, X.,

Karimova, D., Kiers, H. A. L., Klugkist, I., Kuiper, R. M., Lee, M. D., Leenders, R., Leplaa, H. J., Linde, M., Ly, A., Meijerink-Bosman, M., Moerbeek, M., … Hoijtink, H. (2023). A review of applications of the Bayes factor in psychological research. *Psychological Methods*, *28*(3), 558–579. https://doi.org/10.1037/met0000454

Hogeveen, J., Mullins, T. S., Romero, J. D., Eversole, E., Rogge-Obando, K., Mayer, A. R., & Costa, V. D. (2022). The neurocomputational bases of explore-exploit decision-making. *Neuron*, *110*(11), 1869–1879.e5. https://doi.org/10.1016/j.neuron.2022.03.014

Hoijtink, H., Kooten, P. van, & Hulsker, K. (2016). Why bayesian psychologists should change the way they use the bayes factor. *Multivariate Behavioral Research*, *51*(1), 2–10.

Izmirlian, G., Sirota, L. A., Berger, V. W., & Kipnis, V. (2025). The fundamentals of multiplicity adjustment in biostatistics. *JNCI Monographs*, *2025*(68), 10–13. https://doi.org/10.1093/jncimonographs/lgae050

Jeffreys, H. (1938). Significance tests when several degrees of freedom arise simultaneously. *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, *165*(921), 161–198. https://doi.org/10.1098/rspa.1938.0052

Jevremov, T., & Pajić, D. (2024). Bayesian method in psychology: A bibliometric analysis. *Current Psychology*, *43*(10), 8644–8654.

Kelter, R. (2023). How to choose between different bayesian posterior indices for hypothesis testing in practice. *Multivariate Behavioral Research*, *58*(1), 160–188.

Kruschke, J. K. (2011). Bayesian assessment of null values via parameter estimation and model comparison. *Perspectives on Psychological Science*, *6*(3), 299–312. https://doi.org/10.1177/1745691611406925

Kruschke, J. K. (2018). Rejecting or Accepting Parameter Values in Bayesian Estimation. *Advances in Methods and Practices in Psychological Science*, *1*(2), 270–280. https://doi.org/10.1177/2515245918771304

Lakens, D., Scheel, A. M., & Isager, P. M. (2018). Equivalence testing for psychological research: A tutorial. *Advances in Methods and Practices in Psychological Science*, *1*(2), 259–269.

https://doi.org/10.1177/2515245918770963

Li, J., & Ji, L. (2005). Adjusting multiple testing in multilocus analyses using the eigenvalues of a correlation matrix. *Heredity*, *95*(3), 221–227.

Limbachia, C., Morrow, K., Khibovska, A., Meyer, C., Padmala, S., & Pessoa, L. (2021). Controllability over stressor decreases responses in key threat-related brain areas. *Communications Biology*, *4*(1), 42. https://doi.org/10.1038/s42003-020-01537-5

Lindley, D. V. (1972). Bayesian statistics a review. In *Bayesian statistics* (pp. 1–74). Society for Industrial; Applied Mathematics. https://doi.org/10.1137/1.9781611970654.ch1

Ly, A., Verhagen, J., & Wagenmakers, E.-J. (2016). Harold jeffreys's default bayes factor hypothesis tests: Explanation, extension, and application in psychology. *Journal of Mathematical Psychology*, *72*, 19–32. https://doi.org/https://doi.org/10.1016/j.jmp.2015.06.004

Makowski, D., Ben-Shachar, M. S., Chen, S. H. A., & Lüdecke, D. (2019). Indices of effect existence and significance in the bayesian framework. *Frontiers in Psychology*, *Volume 10 - 2019*. https://doi.org/10.3389/fpsyg.2019.02767

Marsman, M., & Wagenmakers, E.-J. (2017). Three insights from a bayesian interpretation of the one-sided p value. *Educational and Psychological Measurement*, *77*(3), 529–539. https://doi.org/10.1177/0013164416669201

Mathar, D., Erfanian Abdoust, M., Marrenbach, T., Tuzsus, D., & Peters, J. (2022). The catecholamine precursor tyrosine reduces autonomic arousal and decreases decision thresholds in reinforcement learning and temporal discounting. *PLOS Computational Biology*, *18*(12). https://doi.org/10.1371/journal.pcbi.1010785

Morey, R. D., & Rouder, J. N. (2011). Bayes factor approaches for testing interval null hypotheses. *Psychological Methods*, *16*(4), 406–419. https://doi.org/10.1037/a0024377

Pesarin, F., & Salmaso, L. (2010). The permutation testing approach: A review. *Statistica*, *70*(4), 481–509. https://doi.org/10.6092/issn.1973-2201/3599

Pfadt, J. M., Bartoš, F., Godmann, H. R., Waaijers, M., Groot, L., Heo, I., et al. (2025, October

30). *A methodological metamorphosis: The rapid rise of bayesian inference and open science practices in psychology*. OSF Preprints. https://doi.org/10.31234/osf.io/ck3js_v1

Pruzek, R. M. (2016). *An introduction to bayesian inference and its applications* (pp. 305–334). Routledge.

Rozeboom, W. W. (1960). The fallacy of the null-hypothesis significance test. *Psychological Bulletin*, *57*(5), 416–428. https://doi.org/10.1037/h0042040

Schnell, P. M., Fiecas, M., & Carlin, B. P. (2020). **credsubs** : Multiplicity-adjusted subset identification. *Journal of Statistical Software*, *94*(7). https://doi.org/10.18637/jss.v094.i07

Scott, J. G., & Berger, J. O. (2006). An exploration of aspects of bayesian multiple testing. *Journal of Statistical Planning and Inference*, *136*(7), 2144–2162.

Scott, J. G., & Berger, J. O. (2010). Bayes and empirical-Bayes multiplicity adjustment in the variable-selection problem. *The Annals of Statistics*, *38*(5), 2587–2619. https://doi.org/10.1214/10-AOS792

Smid, S. C., & Winter, S. D. (2020). Dangers of the defaults: A tutorial on the impact of default priors when using bayesian SEM with small samples. *Frontiers in Psychology*, *Volume 11 - 2020*. https://doi.org/10.3389/fpsyg.2020.611963

Tendeiro, J. N., & Kiers, H. A. (2019). A review of issues about null hypothesis bayesian testing. *Psychological Methods*, *24*(6), 774.

Tendeiro, J. N., Kiers, H. A., Hoekstra, R., Wong, T. K., & Morey, R. D. (2024). Diagnosing the misuse of the bayes factor in applied research. *Advances in Methods and Practices in Psychological Science*, *7*(1), 25152459231213371.

Van De Schoot, R., Winter, S. D., Ryan, O., Zondervan-Zwijnenburg, M., & Depaoli, S. (2017). A systematic review of bayesian articles in psychology: The last 25 years. *Psychological Methods*, *22*(2), 217.

Westfall, P. H., Johnson, W. O., & Utts, J. M. (1997). A bayesian perspective on the bonferroni adjustment. *Biometrika*, *84*(2), 419–427.

Westfall, P. H., & Young, S. S. (1993). *Resampling-based multiple testing: Examples and*

methods for p-value adjustment. Wiley.

Zwet, E. van, & Cator, E. A. (2021). The significance filter, the winner's curse and the need to shrink. *Statistica Neerlandica*, *75*(4), 437–452. https://doi.org/10.1111/stan.12241

Zwet, E. van, & Gelman, A. (2022). A proposal for informative default priors scaled by the standard error of estimates. *The American Statistician*, *76*(1), 1–9. https://doi.org/10.1080/00031305.2021.1938225